

## **BACKLOG BUDDY ? STUDY MATERIAL**

Strength of Materials ? Unit-wise Lecture Notes

Subject Code: ME302 Semester: 3

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Condensed notes covering all five units: Stress and Strain, Shear Force and Bending Moment Diagrams, Bending and Shear Stresses, Torsion of Shafts, Columns and Struts.

### **Unit 1: Stress and Strain**

- Definition, key concepts and significance of Stress and Strain
- Important diagrams / derivations / algorithms for Stress and Strain
- Solved examples and previous-year questions on Stress and Strain

### **Unit 2: Shear Force and Bending Moment Diagrams**

- Definition, key concepts and significance of Shear Force and Bending Moment Diagrams
- Important diagrams / derivations / algorithms for Shear Force and Bending Moment Diagrams
- Solved examples and previous-year questions on Shear Force and Bending Moment Diagrams

### **Unit 3: Bending and Shear Stresses**

- Definition, key concepts and significance of Bending and Shear Stresses
- Important diagrams / derivations / algorithms for Bending and Shear Stresses
- Solved examples and previous-year questions on Bending and Shear Stresses

### **Unit 4: Torsion of Shafts**

- Definition, key concepts and significance of Torsion of Shafts
- Important diagrams / derivations / algorithms for Torsion of Shafts
- Solved examples and previous-year questions on Torsion of Shafts

### **Unit 5: Columns and Struts**

- Definition, key concepts and significance of Columns and Struts
- Important diagrams / derivations / algorithms for Columns and Struts
- Solved examples and previous-year questions on Columns and Struts

Exam Tips:

- \* Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- \* Solve at least 3 previous year papers under timed conditions before the exam.
- \* Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- \* Revise definitions, diagrams and standard derivations ? they earn easy marks.
- \* Practise numerical problems from every unit; papers usually include 2?3 compulsory

## **problems.**

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.